School of Behavioral and Brain Sciences

Neuroscience (BS)

Neuroscience is the multidisciplinary study of nervous system function that draws on recent advances in cell and molecular biology, biochemistry, biophysics, computer, behavioral and cognitive sciences. It examines the brain, spinal cord, and peripheral nervous system's global and nanoscale biochemistry, its complex and extensively networked morphology, and its remarkably adaptive physiology. The field considers neuronal development from early embryology through advanced senescence. Further, neurobiologists examine neuronal plasticity from the level of single proteins, of individual neurons, up through the level of networks or systems of cells, and up to complete behaving organisms. Neuroscience studies the regulation and expression of behavior, the impact of that behavior on neurons, and the complex interactions of multiple neuronal systems that underlie the emergence of cognitive and sensory functions. The Neuroscience program at UT Dallas provides students with opportunities to focus on the brain, spinal cord, and peripheral neurons from a systems and cellular-level perspective, drawing on behavioral and cognitive expertise combined with smaller-scale neuronal and molecular analyses. It allows undergraduates extensive interactions with working neuroscientists who use the latest experimental techniques.

The Neuroscience program is designed to prepare students for admission to medical, dental or allied-health graduate schools (through the Medical Neuroscience track), for admission to research-based doctoral training programs (through the Research Neuroscience track), or for careers in biomedical science, industry, and allied health fields (through the Industrial Neuroscience track). Required courses within the Medical track include the approved pre-medical curriculum and offer a popular alternative to other pre-health majors. Students who wish to continue their education in the fields of medicine, dentistry or allied health professions should also register with the Health Professions Advising Center during their first semester.

Students can choose between three program tracks: Medical Neuroscience Track, Research Neuroscience Track, or Industrial Neuroscience Track to meet their career training needs. All tracks share a Core Curriculum and Neuroscience major requirements that can be completed in a minimum of 82 semester credit hours. Different coursework is required within the different tracks, with the total number of credit hours required for graduation the same for all three tracks.

Bachelor of Science in Neuroscience

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> bbs-neuroscience-bs

Professors: Gregory Dussor, Michael P. Kilgard, Theodore Price, Robert L. Rennaker II, Steven Small, Ana Solodkin

Associate Professors: Michael Burton, Benedict Kolber, Sven Kroener, Christa McIntyre Rodriguez

Assistant Professors: Andrew Eagle, Crystal Engineer, Zirong Gu, Lena Nguyen, Puja Parekh, Millie Rincon-Cortes, Katelyn Sadler, Diana Tavares Ferreira, Catherine Thorn

Professor of Instruction: Van Miller

Associate Professor of Instruction: Steven McWilliams, Siham Raboune, Rukhsana Sultana, Anna Taylor

Assistant Professors of Instruction: Rafael Granja-Vasquez, Faisal Jahangiri, William Marks, Cecily Oleksiak, Amy Zwierzchowski-Zarate

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from Communication Core courses (see advisor)

Mathematics: 3 semester credit hours

MATH 2414 Integral Calculus[1]

or MATH 2417 Calculus I[1]

Or select any 3 semester credit hours from Mathematics Core courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

BIOL 2311 Introduction to Modern Biology I[1]

CHEM 1311 General Chemistry I[1]

Or select 6 semester credit hours from Life and Physical Sciences Core courses (see NSC advisor)

Language, Philosophy and Culture (040): 3 semester credit hours

Select 3 semester credit hours from Language, Philosophy and Culture Core courses (see advisor)

Creative Arts: 3 semester credit hours

Select 3 semester credit hours from Creative Arts Core courses (see advisor)

American History: 6 semester credit hours

Select 6 semester credit hours from American History Core courses (see advisor)

Government/Political Science: 6 semester credit hours

Select 6 semester credit hours from Government/Political Science Core courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

PSY 2301 Introduction to Psychology[1]

Or select any 3 semester credit hours from Social and Behavioral Sciences Core courses (see advisor)

Component Area Option: 6 semester credit hours

PSY 2317 Statistics for Psychology[1]

or STAT 1342 Statistical Decision Making[1]

or STAT 2332 Introductory Statistics for Life Sciences[1]

CHEM 1312 General Chemistry II[1]

Or select any 6 semester credit hours from Component Area Option Core courses (see advisor)

II. Major Requirements: 27 semester credit hours

Major Preparatory Courses: 25 semester credit hours (6 semester credit hours beyond Core Curriculum)[1]

All of the following:

BBSU 1100 First Year Seminar

BIOL 2111 Introduction to Modern Biology Workshop I

BIOL 2281 Introductory Biology Laboratory

BIOL 2311 Introduction to Modern Biology I[1]

CHEM 1111 General Chemistry Laboratory I

CHEM 1311 General Chemistry I[1]

CHEM 1112 General Chemistry Laboratory II

CHEM 1312 General Chemistry II[1]

MATH 2414 Integral Calculus[1]

or MATH 2417 Calculus I[1]

PSY 2301 Introduction to Psychology[1]

PSY 2317 Statistics for Psychology[1]

or STAT 1342 Statistical Decision Making[1]

or STAT 2332 Introductory Statistics for Life Sciences[1]

Major Core Courses required for all Tracks: 21 semester credit hours

All of the following:

NSC 3361 Introduction to Neuroscience

NSC 4352 Cellular Neuroscience

NSC 4353 Neuroscience Laboratory Methods

NSC 4354 Integrative Neuroscience

NSC 4356 Neurophysiology

NSC 4363 Neuropharmacology

NSC 4366 Neuroanatomy

III. Tracks - 51 semester credit hours

Neuroscience majors select one of three Tracks:

- Medical Neuroscience (12 semester credit hours, plus 28 semester credit hours of pre-med courses, plus 11 semester credit hours of free electives)
- Research Neuroscience (12 semester credit hours, plus 28 semester credit hours of pre-graduate courses, plus 11 semester credit hours of free electives)

- Industrial Neuroscience (22 semester credit hours, plus 29 semester credit hours of free electives)

Track 1: Medical Neuroscience

Choose four courses (12 semester credit hours) from the following:

NSC 4310 Principles of Pediatric Neurology

NSC 4320 Sleep and Sleep Disorders

NSC 4330 Fundamentals of EEG and ECoG

NSC 4350 Medical Neuropathology

NSC 4351 Medical Neuroscience

NSC 4358 Neuroscience of Pain

NSC 4362 Molecular Neuroscience

NSC 4364 Journey into Medicine

NSC 4372 Neuroimmunology

NSC 4373 Sensory Neuroscience

NSC 4377 Neurogenetics

NSC 4378 Neurotoxicology

NSC 4380 Neurology of Cognition

NSC 4381 Health Disparities in Neuroscience

NSC 4382 Neurobiology of Emotion

NSC 4383 Human Neurophysiology Lab

NSC 4387 Neuropathology

NSC 4388 Medical Physiology

NSC 4397 Thesis Research

NSC 4V75 Honors Seminar

NSC 4V98 Directed Research[2]

NSC 4V99 Independent Study[3]

Required Pre-medical Basic Biology, Chemistry and Physics (28 semester credit hours)

BIOL 2112 Introduction to Modern Biology Workshop II

BIOL 2312 Introduction to Modern Biology II

BIOL 3461 Biochemistry I

CHEM 2233 Introductory Organic Chemistry Laboratory

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

PHYS 1301 College Physics I

PHYS 1101 College Physics Laboratory I

or PHYS 2125 Physics Laboratory I

PHYS 1302 College Physics II

PHYS 1102 College Physics Laboratory II

or PHYS 2126 Physics Laboratory II

An additional 4 semester credit hours selected with your advisor from NSC or other fields. See advisor for additional guidance.

Elective Requirements: 11 semester credit hours of free electives.

At least 11 semester credit hours of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Track 2: Research Neuroscience

Choose four courses (12 semester credit hours) from the following:

NSC 4320 Sleep and Sleep Disorders

NSC 4330 Fundamentals of EEG and ECoG

NSC 4355 Advanced Neuroscience Laboratory

NSC 4357 Neurobiology of Learning and Memory

NSC 4358 Neuroscience of Pain

NSC 4359 Cognitive Neuroscience

NSC 4362 Molecular Neuroscience

NSC 4367 Developmental Neurobiology

NSC 4371 Neural Plasticity

NSC 4372 Neuroimmunology

NSC 4373 Sensory Neuroscience

NSC 4374 Neuroplasticity in Disorders of the Nervous System

NSC 4376 Neurobiology of Stress

NSC 4377 Neurogenetics

NSC 4380 Neurology of Cognition

NSC 4381 Health Disparities in Neuroscience

NSC 4382 Neurobiology of Emotion

NSC 4383 Human Neurophysiology Lab

NSC 4391 Writing and Independent Study

NSC 4397 Thesis Research

NSC 4V75 Honors Seminar

NSC 4V98 Directed Research

NSC 4V99 Independent Study

Required pre-graduate courses, basic Biology, Chemistry and Physics (28 semester credit hours)

BIOL 2112 Introduction to Modern Biology Workshop II

BIOL 2312 Introduction to Modern Biology II

BIOL 3461 Biochemistry I

CHEM 2233 Introductory Organic Chemistry Laboratory

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

PHYS 1301 College Physics I

PHYS 1101 College Physics Laboratory I

or PHYS 2125 Physics Laboratory I

PHYS 1302 College Physics II

PHYS 1102 College Physics Laboratory II

or PHYS 2126 Physics Laboratory II

An additional 4 semester credit hours selected with your advisor from NSC or other fields. See advisor for additional guidance.

Elective Requirements: 11 semester credit hours of free electives.

At least 11 semester credit hours of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Track 3: Industrial Neuroscience

Required coursework: 16 semester credit hours

NSC 4360 Introduction to Entrepreneurship in Neuroscience

NSC 4361 Pathway Into Industrial Neuroscience

NSC 4193 Internship Preparation

Select 3 additional courses: 9 semester credit hours from this list:

NSC 4391 Writing and Independent Study

NSC 4394 Internship in Neuroscience

NSC 4395 Internship in Neuroscience II

NSC 4397 Thesis Research

NSC 4V95 Externship in Neuroscience

NSC 4V99 Independent Study

Additional Emphasis: 6 semester credit hours

Choose from one of the following two-course groups for additional emphasis:

Entrepreneurship

ENTP 3301 Innovation and Entrepreneurship

ENTP 4311 Entrepreneurial Strategy and Business Models

Healthcare Management

HMGT 3301 Healthcare Management

HMGT 3310 Healthcare Regulation and Economics

Marketing

MKT 3300 Principles of Marketing

MKT 3330 Fundamentals of Professional Selling

Elective Requirements: 29 semester credit hours

29 semester credit hours of free elective coursework, selected with your advisor, from NSC or other fields.

Electives are of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Fast Track Baccalaureate/Master's Degrees

UT Dallas undergraduate students with strong academic records who intend to pursue a master's degree in Applied Cognition and Neuroscience at UT Dallas may consider an accelerated undergraduate-graduate plan of study. If accepted into the program, students may take up to 15 semester credit hours of graduate courses that may be used to complete the baccalaureate degree and also satisfy requirements for the master's degree. Students must maintain a 3.000 grade point average and earn grades of B or better in graduate courses taken.

Students should apply for Fast Track admission in the semester they reach 90 semester credit hours. To qualify for application, undergraduate students must have completed at least 18 semester credit hours in major core courses at UT Dallas. To be eligible for Fast Track admission, students must have completed at least 90 semester credit hours toward a baccalaureate degree, completed a minimum of 36 hours of general education core curriculum classes, and meet program admission requirements. Apply to the Fast Track program through the Applied Cognition and Neuroscience Program Office. Students should consult with a graduate advisor regarding admissions criteria and plans of study.

Minors

Students must complete 18 semester credit hours including 9 required semester credit hours of foundation coursework and 9 semester credit hours of guided electives. At least 12 semester credit hours must be upper-division courses, of which at least 9 semester credit hours must have been completed at UT Dallas.

Minor in Neuroscience

18 semester credit hours

Students who are not majoring in Neuroscience may minor in Neuroscience. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average).

Students should take 9 semester credit hours (3 courses) from the Neuroscience Core courses:

NSC 3361 Introduction to Neuroscience

NSC 4352 Cellular Neuroscience

NSC 4354 Integrative Neuroscience

NSC 4356 Neurophysiology

NSC 4363 Neuropharmacology

NSC 4366 Neuroanatomy

Plus 9 semester credit hours (3 courses) from one of the Neuroscience career tracks:

Medical Neuroscience (9 semester credit hours)

NSC 4350 Medical Neuropathology

NSC 4351 Medical Neuroscience

NSC 4358 Neuroscience of Pain

NSC 4362 Molecular Neuroscience

NSC 4364 Journey into Medicine

NSC 4372 Neuroimmunology

NSC 4373 Sensory Neuroscience

NSC 4382 Neurobiology of Emotion

Research Neuroscience (9 semester credit hours)

NSC 4353 Neuroscience Laboratory Methods

NSC 4357 Neurobiology of Learning and Memory

NSC 4358 Neuroscience of Pain

NSC 4362 Molecular Neuroscience

NSC 4367 Developmental Neurobiology

NSC 4371 Neural Plasticity

NSC 4372 Neuroimmunology

NSC 4373 Sensory Neuroscience

NSC 4382 Neurobiology of Emotion

Industrial Neuroscience (9 semester credit hours)

NSC 4360 Introduction to Entrepreneurship in Neuroscience

NSC 4361 Pathway Into Industrial Neuroscience

NSC 4391 Writing and Independent Study

NSC 4193 Internship Preparation

NSC 4394 Internship in Neuroscience

NSC 4395 Internship in Neuroscience II

NSC 4397 Thesis Research

NSC 4V99 Independent Study

1. A required preparatory course that also fulfills a Core Curriculum requirement. Eighteen (18) semester credit hours are counted in Core Curriculum.

2. May be repeated for credit, up to 9 semester credit hours.

3. May be repeated for credit, up to 6 semester credit hours.

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